$\frac{1}{(1+r)^t}$ → discount factor → measures present value of one dollar received in year t

The longer you have to wait for your money, the lower its present value

TOPIC 3

INTRODUCTION TO TIME VALUE OF MONEY

WHAT WE SHOULD KNOW

- Determine the future value of an investment made today
- Determine the present value of cash to be received at a future date
- Find the return on an investment
- Calculate how long it takes for an investment to reach a desired value

NEW TERMS

- Future Value and Compounding
- Present Value and Discounting
- More about Present and Future Values
- Annuity and Annuity Due

CHALLENGES FOR A MANAGER

- How to determine the value of cash flow expected in future?
- How to allocate cash today for future liabilities?
- How to determine best projects based on expected future cash flows?

BASIC DEFINITIONS

- Present Value – earlier money on a time line
- Future Value – later money on a time line
- Interest rate – "exchange rate" between earlier money and later money
  - Discount rate
  - Cost of capital
  - Opportunity cost of capital
  - Required return

FUTURE VALUE – EXAMPLE 1

- Suppose you invest $1,000 for one year at 5% per year. What is the future value in one year?

- Suppose you leave the money in for another year. How much will you have two years from now?

FUTURE VALUE – EXAMPLE 1

- Suppose you invest $1,000 for one year at 5% per year. What is the future value in one year?
  - Interest = 1,000(.05) = 50
  - Value in one year = principal + interest = 1,000 + 50 = 1,050
  - Future Value (FV) = 1,000(1 + .05) = 1,050

- Suppose you leave the money in for another year. How much will you have two years from now?
  - FV = 1,000(1.05)(1.05) = 1,000$(1.05)^2$ = 1,102.50

FUTURE VALUE: GENERAL FORMULA

- $FV = PV(1 + r)^t$
  - FV = future value
  - PV = present value
  - r = period interest rate, expressed as a decimal
  - t = number of periods

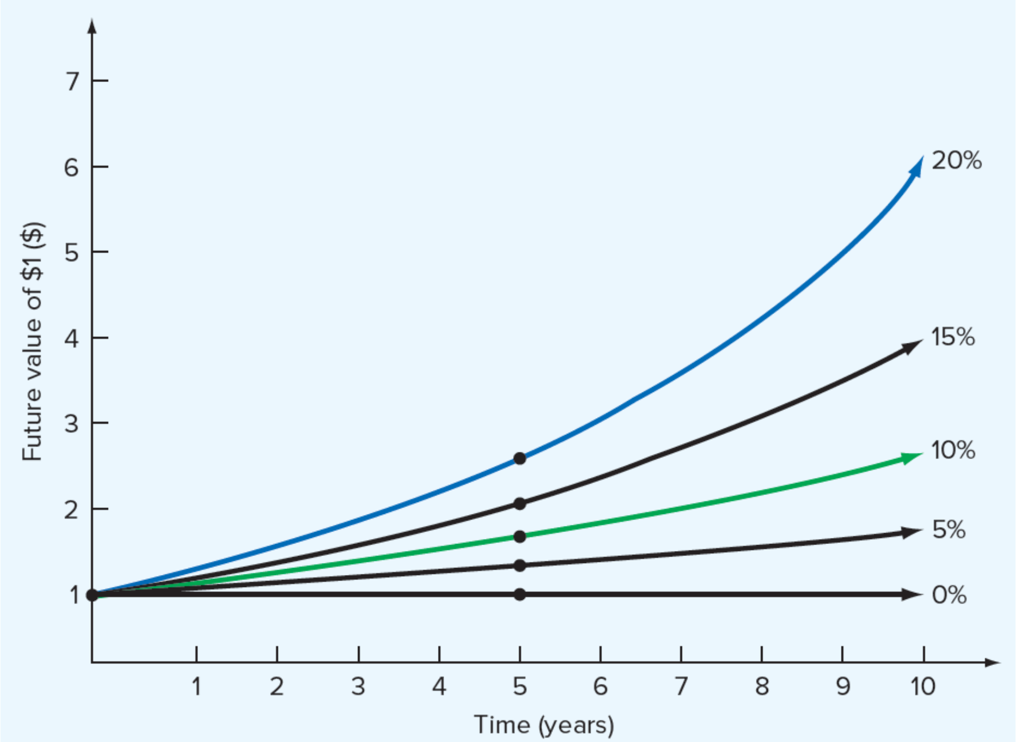
- Future value interest factor = $(1 + r)^t$

EFFECTS OF COMPOUNDING

- Simple interest vs. Compound interest

- Consider the previous example
  - FV with simple interest = 1,000 + 50 + 50 = 1,100
  - FV with compound interest = 1,102.50
  - The extra 2.50 comes from the interest of .05(50) = 2.50 earned on the first interest payment

FUTURE VALUE



$$FVIF_{i,n} = (1 + i)^n$$

Future Value Interest Factors (FVIFs) for $1 at Interest Rate *i* for *n* Periods*

	Interest Rate (*i*)				
End of Period (*n*)	**1%**	**5%**	**6%**	**8%**	**10%**
1	1.010	1.050	1.060	1.080	1.100
2	1.020	1.102	1.124	1.166	1.210
3	1.030	1.158	1.191	1.260	1.331
4	1.041	1.216	1.262	1.360	1.464
5	1.051	**1.276**	1.338	1.469	1.611
8	1.083	1.477	1.594	1.851	2.144
9	1.094	1.551	1.689	**1.999**	2.358
10	1.105	**1.629**	1.791	2.159	2.594
20	1.220	2.653	**3.207**	4.661	6.728
25	1.282	3.386	4.292	6.848	10.835

$$FV_n = PV_0(FVIF_{i,n})$$

FUTURE VALUE – PROBLEM

- Suppose you invest the $1,000 from the previous example for 5 years. How much would you have?

FUTURE VALUE – PROBLEM

- Suppose you invest the $1,000 from the previous example for 5 years. How much would you have?
  - FV = 1,276.28

- The effect of compounding is small for a small number of periods, but increases as the number of periods increases. (Simple interest would have a future value of $1,250, for a difference of $26.28.)

FUTURE VALUE – PROBLEM

- Suppose you had a relative deposit $10 at 5.5% interest 200 years ago. How much would the investment be worth today?

- What is the effect of compounding?
  - Simple interest ???

FUTURE VALUE – PROBLEM

- Suppose you had a relative deposit $10 at 5.5% interest 200 years ago. How much would the investment be worth today?
  - FV = 447,189.84

- What is the effect of compounding?
  - Simple interest = 10 + 200(10)(.055) = 120.00
  - Compounding added $447,069.84 to the value of the investment

FUTURE VALUE AS A GENERAL GROWTH FORMULA

- Suppose your company expects to increase unit sales of widgets by 15% per year for the next 5 years. If you sell 3 million widgets in the current year, how many widgets do you expect to sell in the fifth year?

FUTURE VALUE AS A GENERAL GROWTH FORMULA

- Suppose your company expects to increase unit sales of cellphones by 15% per year for the next 5 years. If you sell 3 million widgets in the current year, how many widgets do you expect to sell in the fifth year?

- Formula: $FV = 3{,}000{,}000(1.15)^5 =$
  - $= 3{,}000{,}000(2.011357187) = 6{,}034{,}072$

  - FV = 6,034,072 units

EXAMPLE

- You've located an investment that pays 12 percent per year. That rate sounds good to you, so you invest $400. How much will you have in three years?
- How much will you have in seven years?
- At the end of seven years, how much interest will you have earned? How much of that interest results from compounding?

SOLUTION

$$(1 + r)^t = 1.12^3 = 1.4049$$

Your \$400 thus grows to:

$$\$400 \times 1.4049 = \$561.97$$

After seven years, you will have:

$$\$400 \times 1.12^7 = \$400 \times 2.2107 = \$884.27$$

Thus, you will more than double your money over seven years.

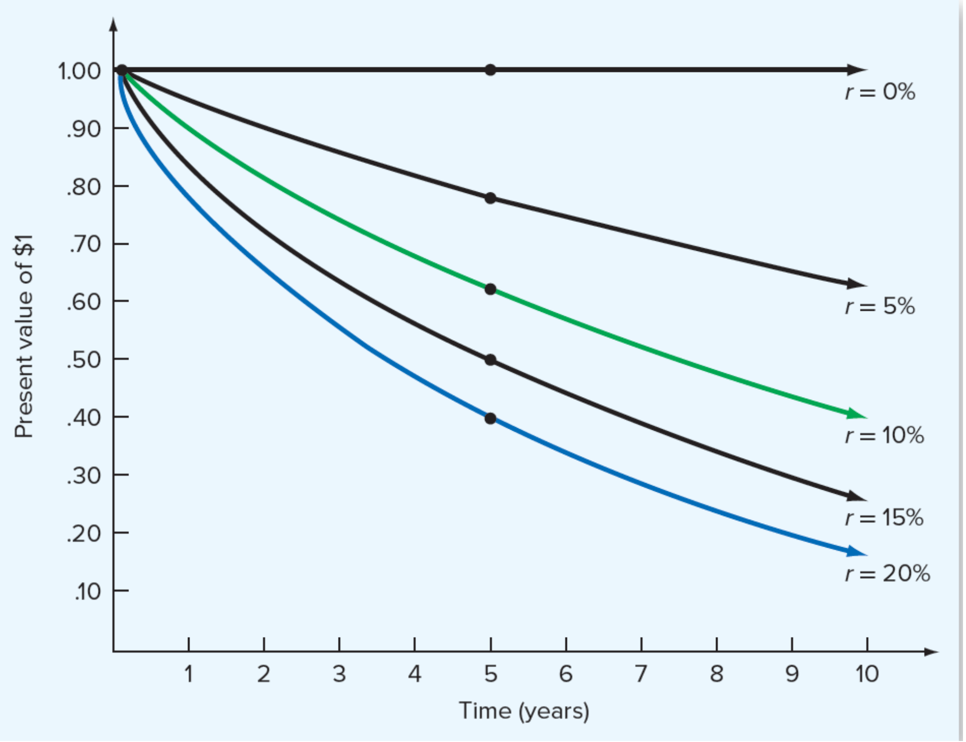
Because you invested \$400, the interest in the \$884.27 future value is \$884.27 − 400 = \$484.27.
At 12 percent, your \$400 investment earns \$400 × 0.12 = \$48 in simple interest every year. Over seven years, the simple interest thus totals 7 × \$48 = \$336.

The other \$484.27 − 336 = \$148.27 is from compounding.

PRESENT VALUE

- How much do I have to invest today to have some amount in the future?
  - $FV = PV(1 + r)^t$
  - Rearrange to solve for $PV = FV / (1 + r)^t$
- When we talk about discounting, we mean finding the present value of some future amount.
- When we talk about the "value" of something, we are talking about the present value unless we specifically indicate that we want the future value.

PRESENT VALUE AND DISCOUNT RATE



Present Value Interest Factors (PVIFs) for $1 at Interest Rate i for *n* Periods

	Interest Rate (*i*)					
End of Period (*n*)	**1%**	**5%**	**6%**	**8%**	**10%**	**13%**
1	0.990	0.952	0.943	0.926	0.909	0.885
2	0.980	0.907	0.890	0.857	0.826	0.783
3	0.971	0.864	0.840	0.794	0.751	0.693
4	0.961	**0.823**	**0.792**	0.735	0.683	0.613
5	0.951	0.784	0.747	0.681	0.621	0.543
8	0.923	0.677	0.627	0.540	0.467	0.376
10	0.905	0.614	0.558	0.463	0.386	0.295
20	0.820	0.377	0.312	0.215	**0.149**	0.087
25	0.780	0.295	0.233	0.146	0.092	0.047

$$PV_0 = FV_n(PVIF_{i,n})$$

$$PVIF_{i,n} = \frac{1}{(1+i)^n}$$

PRESENT VALUE –EXAMPLE 1

- Suppose you need \$10,000 in one year for the down payment on a new car. If you can earn 7% annually, how much do you need to invest today?

- $PV = 10{,}000 / (1.07)^1 = 9{,}345.79$

PRESENT VALUE – EXAMPLE 2

- You want to begin saving for your daughter's college education and you estimate that she will need $150,000 in 17 years. If you feel confident that you can earn 8% per year, how much do you need to invest today?
  - N = 17; I/Y = 8; FV = 150,000
  - PV = 40,540.34

PRESENT VALUE – PROBLEM

- Your parents set up a saving fund for you 10 years ago that is now worth $19,671.51. If the fund earned 7% per year, how much did your parents invest?

PRESENT VALUE – PROBLEM

- Your parents set up a trust fund for you 10 years ago that is now worth $19,671.51. If the fund earned 7% per year, how much did your parents invest?

  - PV = 10,000

PRESENT VALUE – IMPORTANT RELATIONSHIP I

- For a given interest rate – the longer the time period, the lower the present value
  - What is the present value of $500 to be received in 5 years? 10 years? The discount rate is 10%
  - PV = 310.46
  - PV = 192.77

PRESENT VALUE – IMPORTANT RELATIONSHIP II

- For a given time period – the higher the interest rate, the smaller the present value
  - What is the present value of $500 received in 5 years if the interest rate is 10%? 15%?
    - PV = 310.46
    - PV = 248.59

OTHER CASES

Present Value of an Uneven Cash Flow Stream

Year	Cash Flow Rs.	$PVIF_{12\%,n}$	Present Value of Individual Cash Flow
1	1,000	0.893	893
2	2,000	0.797	1,594
3	2,000	0.712	1,424
4	3,000	0.636	1,908
5	3,000	0.567	1,701
6	4,000	0.507	2,028
7	4,000	0.452	1,808
8	5,000	0.404	2,020
Present Value of the Cash Flow Stream			13,376

DISCOUNT RATE – EXAMPLE 1

- You are looking at an investment that will pay $1,200 in 5 years if you invest $1,000 today. What is the implied rate of interest?
  - $r = (1{,}200 / 1{,}000)^{1/5} - 1 = .03714 = 3.714\%$

DISCOUNT RATE – PROBLEM

- Suppose you are offered an investment that will allow you to double your money in 6 years. You have $10,000 to invest. What is the implied rate of interest?

DISCOUNT RATE – PROBLEM

- Suppose you are offered an investment that will allow you to double your money in 6 years. You have $10,000 to invest. What is the implied rate of interest?

  - N = 6
  - PV = -10,000
  - FV = 20,000
  - I/Y = 12.25%

DISCOUNT RATE – PROBLEM

- Suppose you have a 1-year old son and you want to provide $75,000 in 17 years towards his college education.
  - You currently have $5,000 to invest.
  - What interest rate must you earn to have the $75,000 when you need it?

DISCOUNT RATE – PROBLEM

- Suppose you have a 1-year old son and you want to provide $75,000 in 17 years towards his college education.
  - You currently have $5,000 to invest.
  - What interest rate must you earn to have the $75,000 when you need it?

  - I/Y = 17.27%

FINDING THE NUMBER OF PERIODS

- Start with the basic equation and solve for t (remember your logs).
  - $FV = PV(1 + r)^t$
  - $t = \ln(FV / PV) / \ln(1 + r)$

NUMBER OF PERIODS – EXAMPLE 1

- You want to purchase a new car, and you are willing to pay $20,000.
  - If you can invest at 10% per year and you currently have $15,000, how long will it be before you have enough money to pay cash for the car?

- N = 3.02 years
- Formula: t = ln(20,000 / 15,000) / ln(1.1) = 3.02 years

NUMBER OF PERIODS – EXAMPLE 2

- Suppose you want to buy a new house.
  - You currently have $15,000, and you figure you need to have a 10% down payment plus an additional 5% of the loan amount for closing costs.
  - Assume the type of house you want will cost about $150,000 and you can earn 7.5% per year.
  - How long will it be before you have enough money for the down payment and closing costs?

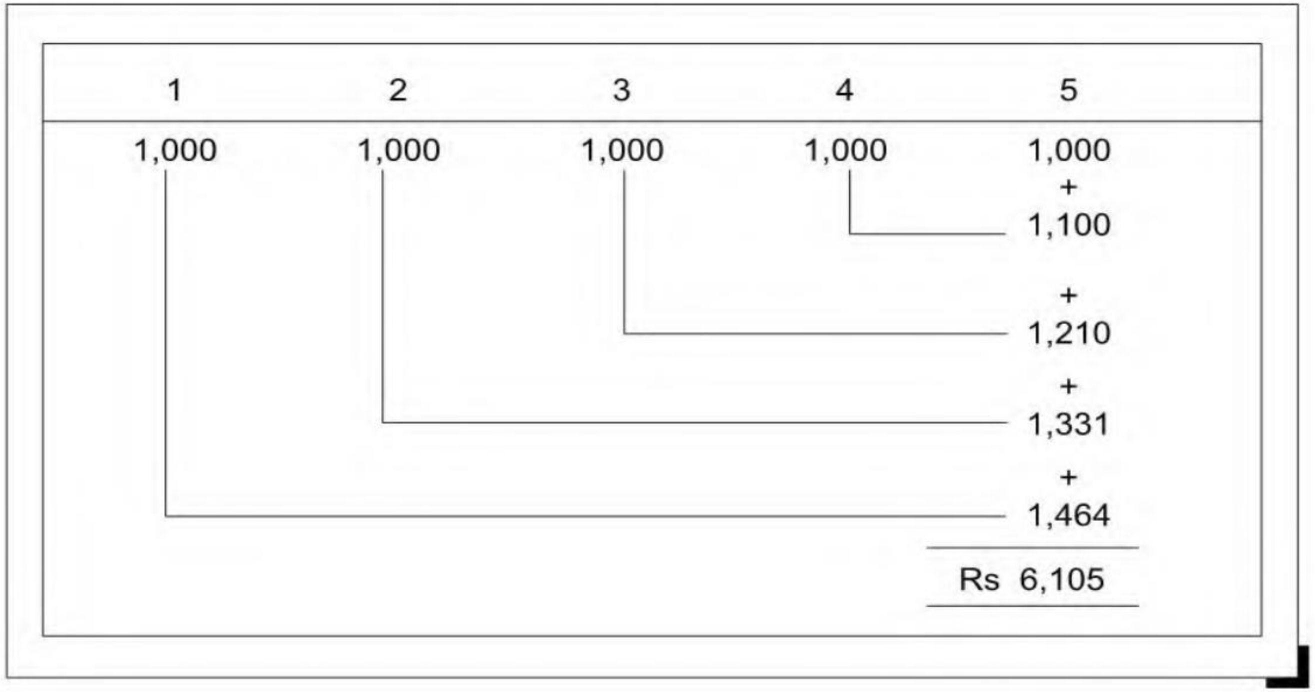
ANNUITY AND ANNUITY DUE

- Future value of an Annuity
- Future value of an Annuity Due
- Present value of an Annuity
- Present value of an Annuity Due

FUTURE VALUE OF AN ANNUITY

- Suppose you deposit Rs.1,000 annually in a bank for 5 years and your deposits earn a compound interest rate of 10 percent. What will be the value of this series of deposits (an annuity) at the end of 5 years?

SOLUTION



$$\text{Rs.}1{,}000(1.10)^4 + \text{Rs.}1{,}000(1.10)^3 + \text{Rs.}1{,}000(1.10)^2 + \text{Rs.}1{,}000(1.10) + \text{Rs.}1{,}000$$

$$= \text{Rs.}1{,}000(1.464) + \text{Rs.}1{,}000(1.331) + \text{Rs.}1{,}000(1.21) + \text{Rs.}1{,}000(1.10) + \text{Rs.}1{,}000$$

$$= \text{Rs.}6{,}105$$

EXAMPLE: HOW LONG SHOULD YOU WAIT

- You want to take up a trip to the Europe with a friend which costs Rs.10,00,000 - the cost is expected to remain unchanged in nominal terms. You can save annually Rs.50,000 to fulfill your desire. How long will you have to wait if your savings earn an interest of 12 percent?

SOLUTION

The future value of an annuity of Rs.50,000 that earns 12 percent is equated to Rs.10,00,000.

$$50{,}000 \times \text{FVIFA}_{n=?,12\%} = 1{,}000{,}000$$

$$50{,}000 \times \left[\frac{1.12^n - 1}{0.12}\right] = 1{,}000{,}000$$

$$1.12^n - 1 = \frac{1{,}000{,}000}{50{,}000} \times 0.12 = 2.4$$

$$1.12^n = 2.4 + 1 = 3.4$$

$$n \log 1.12 = \log 3.4$$

$$n \times 0.0492 = 0.5315$$

$$n = \frac{0.5315}{0.0492} = 10.8 \text{ years}$$

You will have to wait for about 11 years.

FORMULA

$$FVA_n = A(1+r)^{n-1} + A(1+r)^{n-2} + \cdots + A$$
$$= A[(1+r)^n - 1] / r$$

Derivation of formula

The future value of an annuity is:

$$FVA_n = A(1+r)^{n-1} + A(1+r)^{n-2} + \cdots + A(1+r) + A \quad (1)$$

Multiplying both the sides of (1) by $(1+r)$ gives:

$$FVA_n(1+r) = A(1+r)^n + A(1+r)^{n-1} + \cdots + A(1+r)^2 + A(1+r) \quad (2)$$

Subtracting (1) from (2) yields:

$$FVA_n\, r = A[(1+r)^n - 1] \quad (3)$$

Dividing both the sides of (3) by r gives:

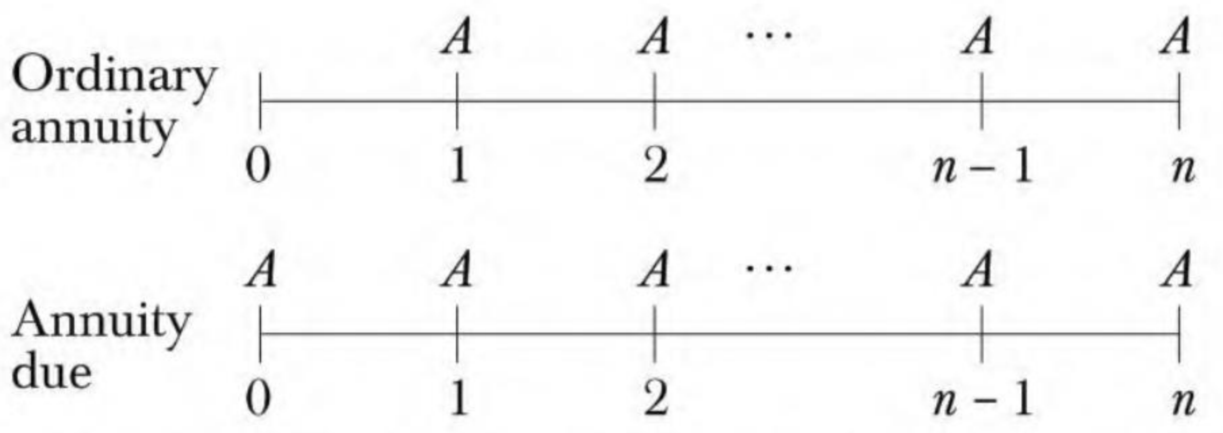
$$FVA_n = A\left[\frac{(1+r)^n - 1}{r}\right] \quad (4)$$

$\mathbf{FVIFA}_{r,n}$

FORMULA

- $FV\ of\ Annuity = A\left[\frac{(1+r)^n - 1}{r}\right]$
- $FV\ of\ Annuity\ Due = A\left[\frac{(1+r)^n - 1}{r}\right]$*(1+r)
- $PV\ of\ Annuity = A\left[\frac{1-(1+r)^{-n}}{r}\right]$
- $PV\ of\ Annuity = A\left[\frac{1-(1+r)^{-n}}{r}\right]$*(1+r)

ANNUITY VS ANNUITY DUE



Annuity due value = Ordinary annuity value × $(1 + r)$

MULTIPERIODS COMPOUNDING

- $FV\ of\ Annuity = A\left[\frac{(1+r/m)^{n*m}-1}{r/m}\right]$
- $PV\ of\ Annuity = A\left[\frac{1-\ (1+r/m)^{-n*m}}{r/m}\right]$
- Present value of a perpetuity= A/r
- Present value of a perpetuity with constant growth= A/(r-g)
- $Effective\ Annual\ rate(EAR) = A = \left[1 + \frac{r}{m}\right]^{n*m} - 1$

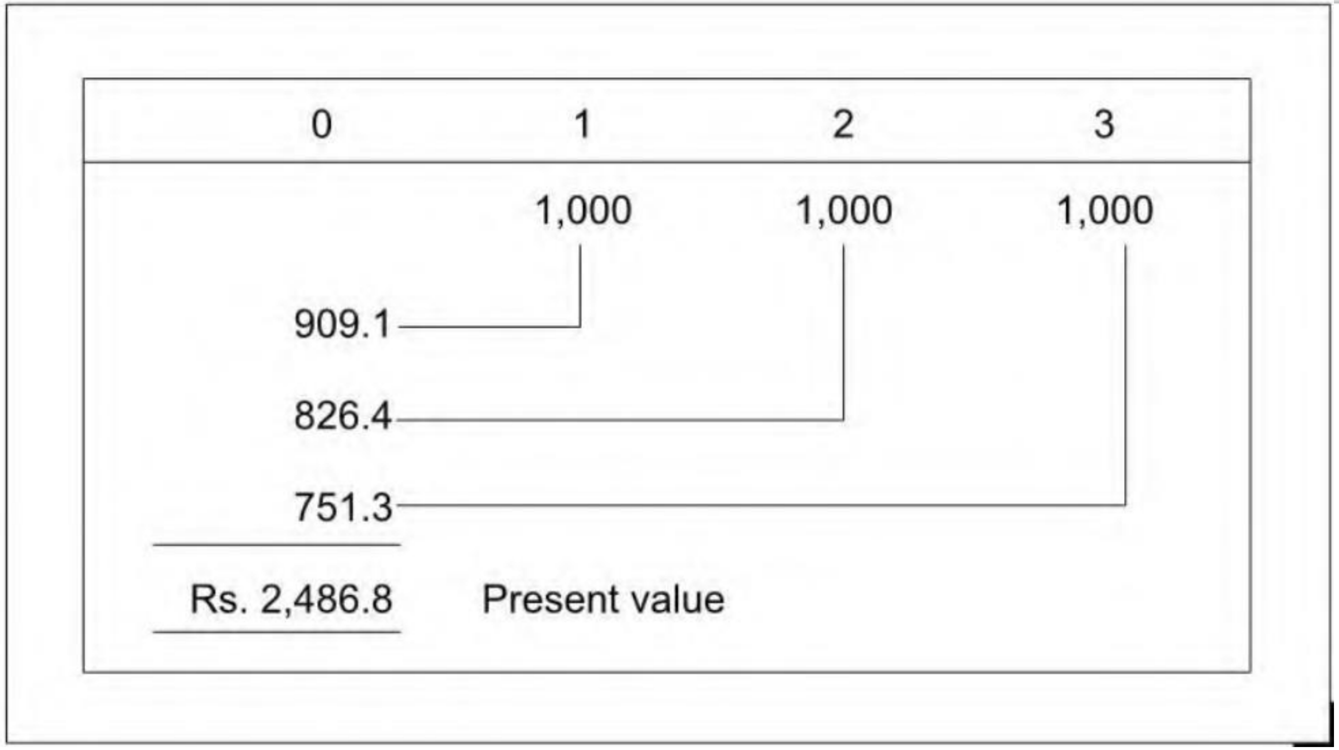
COMPREHENSIVE PROBLEM

- Suppose you are taking a loan of Rs 10 lacs from SBI bank at 12% interest rate for 5 years. Compute EMI that you'll be paying towards this loan. What would be total interest amount on this loan?

PRESENT VALUE OF AN ANNUITY

- Suppose you expect to receive Rs. 1,000 annually for 3 years, each receipt occurring at the end of the year. What is the present value of this stream of benefits if the discount rate is 10 percent?

SOLUTION



$$Rs.\ 1{,}000\left[\frac{1}{1.10}\right] + Rs.\ 1{,}000\left[\frac{1}{1.10}\right]^2 + Rs.\ 1{,}000\left[\frac{1}{1.10}\right]^3$$

$$= Rs.1{,}000 \times 0.9091 + Rs.1{,}000 \times 0.8264 + Rs.1{,}000 \times 0.7513$$

$$= Rs.2{,}486.8$$

FORMULA

$$\mathrm{PVA}_n = \frac{A}{(1+r)} + \frac{A}{(1+r)^2} + \cdots + \frac{A}{(1+r)^{n-1}} + \frac{A}{(1+r)^n}$$

$$= A\left[\frac{1}{(1+r)} + \frac{1}{(1+r)^2} + \ldots + \frac{1}{(1+r)^{n-1}} + \frac{1}{(1+r)^n}\right]$$

$$= A\,[\{\,1 - (1/1+r)^n\}/\,r]$$

$$\text{PV of a Growing Annuity} = A\,(1+g)\left[\frac{1 - \dfrac{(1+g)^n}{(1+r)^n}}{r-g}\right]$$

DERIVATION

$$\text{PVA}_n = A\,(1+r)^{-1} + A\,(1+r)^{-2} + \cdots + A\,(1+r)^{-n} \qquad (1)$$

Multiplying both the sides of (1) by (1 + r) gives :

$$\text{PVA}_n\,(1+r) = A + A\,(1+r)^{-1} + \cdots + A\,(1+r)^{1-n} \qquad (2)$$

Subtracting (1) from (2) yields :

$$\text{PVA}_n\,r = A\,[1-(1+r)^{-n}] = A\,\{[(1+r)^n - 1]\,/\,(1+r)^n\} \qquad (3)$$

Dividing both the sides of (3) by r results in :

$$\text{PVA}_n = A\,\{[(1+r)^n - 1]\,/\,r\,(1+r)^n\} = A\,[\{1-(1/1+r)^n\}/\,r \qquad (4)$$

$\text{PVIFA}_{n,r}$

$$\text{PVIFA}_{r,n} = \frac{1-\dfrac{1}{(1+r)^n}}{r}$$

OTHER FORMULA

Present Value of a Perpetuity

$$P_{\infty} = A \times \text{PVIFA}_{r,\infty}$$

$$\text{PVIFA}_{r,\infty} = \frac{1}{r}$$

Growing Perpetuity

$$\text{PV} = \frac{C}{(1+r)} + \frac{C\,(1+g)}{(1+r)^2} + \frac{C\,(1+g)^2}{(1+r)^3} + \cdots + \frac{C\,(1+g)^{n-1}}{(1+r)^n} + \cdots$$

$$\text{PV} = \frac{C}{r-g}$$

EXAMPLE

- You want to borrow Rs. 1,080,000 to buy a flat. You approach a housing finance company which charges 12.5 percent interest. You can pay Rs.180,000 per year toward loan amortisation. What should be the maturity period of the loan?

SOLUTION: PERIOD OF LOAN AMORTISATION

$$180{,}000 \times PVIFA_{n,r} = 1{,}080{,}000$$

$$180{,}000 \times PVIFA_{n=?,\, r=12.5\%} = 1{,}080{,}000$$

$$180{,}000 \left[\frac{1 - \dfrac{1}{(1.125)^n}}{0.125} \right] = 1.080{,}000$$

$$\frac{1 - \dfrac{1}{(1.125)^n}}{0.125} = \frac{1{,}080{,}000}{180{,}000} = 6$$

$$\frac{1}{(1.125)^n} = 0.25$$

$$1.125^n = 4$$

$$n \log 1.125 = \log 4$$

$$n \times 0.0512 = 0.6021$$

$$n = \frac{0.6021}{0.0512} = 11.76 \text{ years}$$

You can perhaps request for a maturity of 12 years.

APPLICATION OF TVM

Determining the Loan Amortisation Schedule

Suppose a firm borrows Rs.1,000,000 at an interest rate of 15 percent and the loan is to be repaid in 5 equal instalments payable at the end of each of the next 5 years.

$$\text{Loan amount} = A \times PVIFA_{n=5,\, r=15\%}$$

$$1{,}000{,}000 = A \times 3.3522$$

$$\text{Hence } A = 298{,}312$$

LOAN AMORTISATION SCHEDULE

Year	Beginning Amount (1)	Annual Instalment (2)	Interest (3)	Principal Repayment (2)-(3) = (4)	Remaining Balance (1)-(4) = (5)
1	1,000,000	298,312	150,000[a]	148,312[b]	851,688
2	851,688	298,312	127,753	170,559	681,129
3	681,129	298,312	102,169	196,143	484,986
4	484,986	298,312	72,748	225,564	259,422
5	259,422	298,312	38,913	259,399	23*

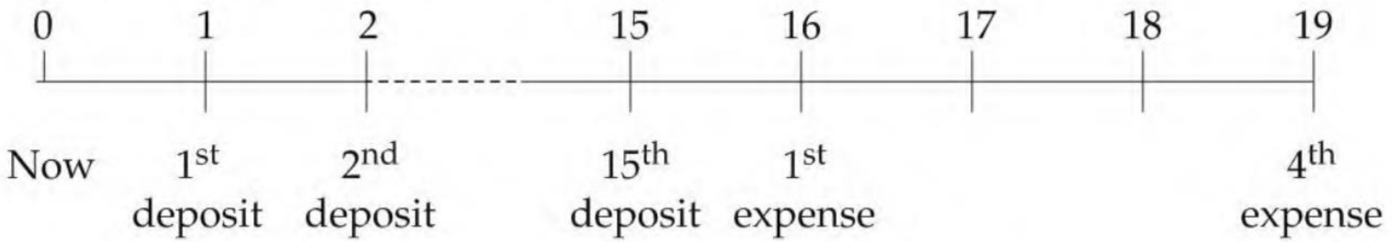
a. Interest is calculated by multiplying the beginning loan balance by the interest rate.
b. Principal repayment is equal to annual instalment minus interest.
* Due to rounding off error a small balance is shown.

OTHER EXAMPLE

- Ravi wants to save for the college education of his son, Deepak. Ravi estimates that the college education expenses will be rupees one million per year for four years when his son reaches college in 16 years - the expenses will be payable at the beginning of the years. He expects the annual interest rate of 8 percent over the next two decades. How much money should he deposit in the bank each year for the next 15 years (assume that the deposit is made at the end of the year) to take care of his son's college education expenses?

SOLUTION

The time line for this problem is as follows:



The present value of college education expenses when his son becomes 15 years old is:

Rs. 1,000,000 × PVIFA (4 years, 8%)

= Rs. 1,000,000 × 3.312 = Rs. 3,312,000

The annual deposit to be made so that the future value of the deposits at the end of 15 years is Rs.3,312,000 is:

$$A = \frac{\text{Rs. } 3{,}312{,}000}{\text{FVIFA(15 years, 8\%)}} = \frac{\text{Rs. } 3{,}312{,}000}{27.152}$$

= Rs. 121,980

Effective versus Stated Rate

$$\text{Effective interest rate} = \left[1 + \frac{\text{Stated annual interest rate}}{m}\right]^m - 1$$

$$\text{Effective interest rate with annual compounding} = \left[1 + \frac{0.12}{1}\right]^1 - 1 = 0.12$$

$$\text{Effective interest rate with semi-annual compounding} = \left[1 + \frac{0.12}{2}\right]^2 - 1 = 0.1236$$

$$\text{Effective interest rate with quarterly compounding} = \left[1 + \frac{0.12}{4}\right]^4 - 1 = 0.1255$$

When compounding becomes continuous

Effective interest rate = $e^r - 1$

QUICK VIEW

Frequency	Stated Interest Rate (%)	m	Formula	Effective Interest Rate (%)
Annual	12	1	0.12	12.00
Semi-annual	12	2	$\left[1+\frac{0.12}{2}\right]^{2}-1$	12.36
Quarterly	12	4	$\left[1+\frac{0.12}{4}\right]^{4}-1$	12.55
Monthly	12	12	$\left[1+\frac{0.12}{12}\right]^{12}-1$	12.68
Weekly	12	52	$\left[1+\frac{0.12}{52}\right]^{52}-1$	12.73
Daily	12	365	$\left[1+\frac{0.12}{365}\right]^{365}-1$	12.75
Continuous	12		$e^{0.12}-1$	12.75

PRINCIPAL AND INTEREST

- Most mortgages are amortizing loans. For example, suppose that you take out a \$250,000 house mortgage from your local savings bank when the interest rate is 12%. The bank requires you to repay the mortgage in equal annual installments over the next 30 years.
- Thus, principal and interest payment will be

$$\text{Annual mortgage payment} = \$250{,}000/30\text{-year annuity factor}$$

$$\text{30-year annuity factor} = \left[\frac{1}{.12} - \frac{1}{.12(1.12)^{30}}\right] = 8.055$$

$$\text{Annual mortgage payment} = 250{,}000/8.055 = \$31{,}036$$

END OF SESSION

TIME VALUE OF MONEY